

Jenkins Distributed Build Lab - Complete Runbook

1. Objective

Deploy a Jenkins controller and SSH agent on separate EC2 instances, validate distributed builds, and document setup, troubleshooting, and cleanup.

2. Architecture

```
GitHub
|
Controller (EC2 t3.small)
|
SSH
|
Agent (EC2 t3.small)
```

3. SSH Authentication Setup

3.1 Generate SSH key on Controller

```
ssh-keygen -t ed25519 -C "jenkins-controller"
# Accept default:
/home/ubuntu/.ssh/id_ed25519

ls -l ~/.ssh
cat ~/.ssh/id_ed25519.pub
```

3.2 Configure Agent authorized_keys

```
mkdir -p ~/.ssh
chmod 700 ~/.ssh
nano ~/.ssh/authorized_keys
# Paste controller's id_ed25519.pub
chmod 600 ~/.ssh/authorized_keys
ls -ld ~/.ssh
ls -l ~/.ssh
```

3.3 Validate SSH

```
ssh ubuntu@<agent-private-ip>
# Login should succeed without password
```

3.4 Jenkins Credentials

Manage Jenkins → Credentials → System → Global Credentials → Add Credentials

```
Kind      : SSH Username with private key
Username  : ubuntu
Private Key: Enter directly
Paste:
cat ~/.ssh/id_ed25519
```

```
ID      : linux-agent-key
Description: SSH Key for Linux Agent
```

4. Create SSH Agent

Create a Permanent Agent and choose 'Launch agents via SSH'. Use the credential 'linux-agent-key'.

5. Verify Agent

```
ps -ef | grep remoting
```

6. Schedule Jobs

```
pipeline {
  agent { label 'linux-agent' }
  stages {
    stage('Verify'){
      steps{
        sh 'hostname'
        sh 'pwd'
      }
    }
  }
}
```

7. Disk Space Warning Investigation

```
df -h /tmp
# Script Console
println System.getProperty("java.io.tmpdir")
```

8. Controller Fix

```
sudo mkdir -p /var/lib/jenkins/tmp
sudo chown jenkins:jenkins /var/lib/jenkins/tmp
sudo chmod 1777 /var/lib/jenkins/tmp

sudo systemctl edit jenkins
[Service]
Environment="JENKINS_JAVA_OPTIONS=-Djava.io.tmpdir=/var/lib/jenkins/tmp"

sudo systemctl daemon-reload
sudo systemctl restart jenkins
```

9. Agent Fix

```
mkdir -p /home/ubuntu/jenkins-agent/tmp
chmod 1777 /home/ubuntu/jenkins-agent/tmp

sudo tee /usr/local/bin/java-jenkins >/dev/null <<'EOF'
#!/bin/bash
exec /usr/bin/java -Djava.io.tmpdir=/home/ubuntu/jenkins-agent/tmp "$@"
EOF
sudo chmod +x /usr/local/bin/java-jenkins

# Node Configuration
JavaPath:
/usr/local/bin/java-jenkins

Reconnect agent

ps -ef | grep remoting
```

10. Validation

```
# Controller Script Console
println System.getProperty("java.io.tmpdir")

# Agent
ps -ef | grep remoting

# Pipeline
```

```
node('linux-agent'){  
  sh 'hostname'  
  sh 'pwd'  
}
```

11. Lessons Learned

Controller and agent are independent JVMs. `java.io.tmpdir` is JVM-specific. Ubuntu tmpfs may trigger false Disk Space Monitor warnings. `JavaPath` wrapper solves SSH agent JVM startup customization.

12. Cleanup

Terminate EC2 instances and retain this runbook for future recreation.